TN SKILL- PROJECT

ON

ket Player Performance Tracking Application : CRICKSTATS

Project work submitted to

THIRUVALLUVAR UNIVERSITY – VELLORE

In partial fulfillment of the requirement

For the award of the degree of

BACHELOR OF SCIENCE IN

SOFTWARE COMPUTER SCIENCE

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User Story:

Courier and logistics companies often struggle to efficiently manage shipments
and deliveries due to reliance on manual processes and disconnected systems.
The absence of a centralized shipment tracking mechanism makes it difficult to
monitor parcel movement in real time, resulting in delayed delivery updates
and poor visibility for customers. Delivery agent assignments are performed
manually, increasing the chances of scheduling conflicts and operational delays. Furthermore, the lack of automated notifications prevents timely communication with customers, while the absence of performance analytics
limits managerial insight into operational efficiency. Managing multiple branches becomes complex without an integrated platform, leading to coordination issues across locations. Collectively, these challenges result in
customer dissatisfaction, reduced productivity, operational inefficiencies, and
ultimately revenue lossf or the organization.

1. I NTRODUCTI ON

1.1 Proj ect Overview:

Courier and logistics companies face challenges in efficiently managing shipments and deliveries due to manual operations and disconnected systems.
These issues lead to delayed updates, poor customer visibility, and operational
inef ficiencies.
The Courier Management System (CMS) is a Salesforce-based cloud solution
that centralizes shipment booking, tracking, delivery agent assignment, automated status updates, billing, and analytics. The system improves operational efficiency, enhances customer satisfaction, and provides real-time
businessinsights.

1.2 Purpose :

The purpose of the Courier Management System (CMS) is to automate and
centralize courier and logistics operations using the Salesforce platform. The
system aims to replace manual and fragmented processes with an integrated
digital solution that enables efficient shipment booking, real-time

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I m trying to	But	Because	Which makes me f eel
PS- 1	Branch Manager	manage shipments ef ficiently	tracking is manual	systems are disconnect ed	stressed
PS- 2	Customer	track parcels	updates are delayed	no centralize d system	uncertain

PS
I am (Customer)
I m trying to
But
Because
Which makesme f eel
PS- 1
Branch Manager
manage shipmentsef ficiently
tracking ismanual
systemsare disconnected
stressed
PS- 2
Customer
track parcels
updatesare delayed
no centralized system
uncertain

2.2 Empathy Map Canvas

User Persona: Branch Manager

Think & Feel: Concerned about delays& errors

Hear: Customer complaints

See: Registers& spreadsheets

Say & Do: Manual f ollow- ups

Pain: Delivery conf licts& missed updates

Gain: Automated tracking & dashboards

2.3 Brainstorming & I dea Prioritization

I deasConsidered

Manual registers

Spreadsheet tracking

Standalone portal

Salesf orce CRM

Final Selection
Salesf orce CRMselected due to:

Automation

Scalability

Real- time visibility

3.REQUI REMENT ANALYSI SPHASE

3.1 Customer Journey Map

Stages:

Booking Shipment Creation Agent Assignment Transit Delivery Update I nvoice Reporting

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR- 1	Customer management
FR- 2	Shipment booking
FR- 3	Delivery agent assignment
FR- 4	Real- time statusupdates
FR- 5	I nvoice generation
FR- 6	Reports& dashboards

FR No
Requirement
FR- 1
Customer management
FR- 2
Shipment booking
FR- 3
Delivery agent assignment
FR- 4
Real- time statusupdates
FR- 5

I nvoice generation
FR- 6
Reports& dashboards
Non- Functional
Requirements

NFR	Description
Usability	Easy Lightning interf ace
Security	Role- based accesscontrol
Reliability	Error- f ree automation
Perf ormance	Fast tracking updates
Availability	24/ 7 cloud access
Scalability	Supportsgrowing branches

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Description
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Easy Lightning interf ace
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Fast tracking updates
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24/ 7 cloud access
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branches

3.3 Data FlowDiagram

Flow:
Customer Input Shipment Creation Validation Agent Assignment Delivery Update I nvoice Reports

3.4 Technology Stack

Layer	Technology
UI	Salesf orce Lightning
Logic	Salesf orce Flows
Database	Salesf orce Obj ects
Automation	Validation Rules, Flows
Security	Profiles, Roles
Reporting	Reports& Dashboards

Layer

Technology
UI
Salesf orce Lightning
Logic
Salesf orce Flows
Database
Salesf orce Obj ects
Automation
Validation Rules,
Flows
Security
Profiles, Roles
Reporting
Reports&
Dashboards

4. PROJECT DESI GNPHASE

4.1 Problem Solution Fit

CMS eliminates fragmented courier operations by automating shipment
tracking, delivery assignment, billing, and reporting within a centralized
Salesf orce platf orm.

4.2 Proposed Solution

Functional Requirements

Parameter	Description
Problem	Manual shipment management
Solution	Salesf orce CMS
I nnovation	Automated tracking & billing
I mpact	Faster deliveries
BusinessModel	Courier services
Scalability	Multi- branch ready

Parameter
Description
Problem
Manual shipment
management
Solution
Salesf orce CMS
I nnovation
Automated tracking & billing
I mpact
Faster deliveries
BusinessModel

Scalability
Multi- branch ready

4.3 Solution Architecture

Components:

Customer Obj ect

Shipment Obj ect

Delivery Agent Obj ect

Branch Obj ect

Delivery Update Obj ect

I nvoice Obj ect

Flows

Reports

Dashboards

5. PROJECT PLANNI NG & SCHEDULI NG

5.1 Proj ect Planning

Agile methodology

Epics Stories Story Points

Sprint- based execution

Velocity calculation used f or estimation

Product Backlog, Sprint Schedule, and Estimation (4 Marks)
Use the belowtemplate to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint- 1	Developer Setup	USN- 1	As a system administrator, I want to create	3	High	Member1

Sprint
Functional Requirement (Epic)
User Story Number
User Story / Task
Story Points
Priority
Team Members
Sprint- 1
Developer Setup
USN- 1
As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Courier Management
System.
3
High
Member1
Sprint- 2
Data Modeling
USN- 2
As an admin, I want to create Customer, Shipment, Delivery Agent, Branch,
and Invoice objects with proper relationships so that courier data is stored and
managed ef ficiently.
5
High
Member2
Sprint- 2
Data Modeling
USN- 3
As a staff member, I want tabs and a Lightning App for all courier modules so
that I can easily navigate and perf orm daily operations.
5
High
Member3
Sprint- 3
Automation
USN- 4
As a system, I want to validate shipment details such as delivery dates, addresses, and mandatory fields so that incorrect or incomplete records are
prevented.
3
High
Member4
Sprint- 3
Automation

As a dispatcher, I want automated flows for shipment status updates and
agent assignment so that deliveries are processed faster without manual
intervention.
3
High
Member4
Sprint- 4
Security
USN- 6
As an administrator, I want role-based access control for managers, agents,
and staf f so that sensitive shipment and billing data issecure.
5
High
Member2
Sprint- 5
Reports
USN- 7
As a branch manager, I want reports showing delivery performance, shipment
status, and revenue so that I can monitor operations and make informed
decisions.
4
High
Member3
Sprint- 6
Dashboards
USN- 8
As a branch manager, I want dashboards showing delivery performance,
shipment status, and revenue so that I can monitor operations and make
inf ormed decisions.
4
Medium
Member4
Proj ect Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned) Planned End	Story Points Completed (as on	Sprint Release Date (Actual)

Designing custom obj ectsand relationships

Creating page layouts& record types

Writing validation rules

Building automation with f lows

Developing reports& dashboards

Setting up security with profiles, permission sets, and sharing

End- to- end application testing

Milestone 1 : Salesf orce Account

I ntroduction :

New to Salesforce and unsure where to start? Curious how real business applications like loan processing and payment tracking can be built on the platf orm? You re in the right place.

Welcome to the Loan Management System proj ect!

Salesforce offers powerful tools that help automate processes, improve accuracy, and manage customer operations with ease. In this project, you ll learn how Salesforce can streamline lending workflows from borrower onboarding to loan approval, EMI tracking, and document verification.

What I sSalesf orce ?

Salesforce is your customer success platform, designed to help you sell, service,
market, analyze, and connect with your customers. Salesforce has everything
you need to run your business from anywhere. Using standard products and
features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

Activity 1: Creating Developer Account

Creating a developer org in salesf orce.

Go to https:// developer.salesf orce.com/ signup

On the sign up f orm, enter the f ollowing details:

3. First name & Last
name
4. Email
1) First name & Last
name
2) Email
3) Job Title: Developer
4) Company: College
Name

Click on sign me up af ter filling these.

Activity 2 : Account Activation

1. Go to the inbox of the email that you used while signing up. Click on th verify
account to activate your account. The email may take 10-30 mins and sometimes2 hours.
2. Click on Verif y Account
3.Give a password and answer a security question and click on change password.
4.Then you will redirect to your salesf orce setup page.

Milestone 2- Obj ect:

What isan obj ect?
Salesforce objects are database tables that permit you to store data that is
specific to an
organization. I t consistsof fields(columns) and records(rows).
Custom Obj ectsCreated:
1. Customer__c stores customer personal and contact information 2. Shipment__c manages parcel details, source, destination, and tracking information 3. Delivery_Agent__c maintains delivery agent profiles and
branch details 4. Branch__c represents courier branch offices 5. Delivery_Update__c stores status updates for each shipment 6. I nvoice_ _ c managesbilling detailsf or delivered shipments
Relationships:
Customer Shipment (Lookup) Shipment Delivery_Agent (Lookup) Delivery_Agent Branch (Lookup) Delivery_Update Shipment (Master- Detail) I nvoice Shipment (Lookup)
Stepsto I mplement :

Activity 1: Create Customer Obj ect

Click Setup Obj ect Manager Create Custom Obj ect

Enter the f ollowing details:

Label: Customer

Plural Label: Customers

Record Name: Customer Name (Text)

Under Optional Features:

Enable AllowReports

Enable Track Field History

I n Search Statuscheck the box AllowSearch

In Object Creation Options check the box Add Notes and Attachments related def ault page layout

Click Save.

Activity 2: Create Shipment Obj ect

Go to Setup Obj ect Manager Create Custom Obj ect

Enter:

Label: Shipment

Plural Label: Shipments

Record Name: Tracking Number (Auto Number)

Example Format: SHP- {00000} Starting Number: 0

Enable:

AllowReports

Track Field History

Click Save

Activity 3: Create Delivery_ Agent Obj ect

Setup Obj ect Manager Create Custom Obj ect

Enter:

Label: Delivery Agent

Plural Label: Delivery Agents

Record Name: Agent Name (Text)

Enable:

AllowReports

Track Field History

Click Save

Activity 4: Create Branch Obj ect

Setup Obj ect Manager Create Custom Obj ect

Enter:

Label: Branch

Plural Label: Branches

Record Name: Branch Name (Text)

Enable:

AllowReports

Click Save.

Activity 5: Create Delivery_ Update Obj ect

Setup Obj ect Manager Create Custom Obj ect

Enter:

Label: Delivery Update

Plural Label: Delivery Updates

Record Name: Update Number (Auto Number)

Example Format: UPD- {0000} Enter Starting number : 0

Enable:

AllowReports

Track Field History

Click Save.

Activity 6: Create I nvoice Obj ect

Setup Obj ect Manager Create Custom Obj ect

Enter:

Label: I nvoice

Plural Label: I nvoices

Record Name: I nvoice Number (Auto Number)

Example Format: I NV- {0000} Enter Starting number: 0

Enable:

AllowReports

Track Field History

Click Save

Milestone 3 - Tabs

Tab:

Tabs in Salesforce help users view the information at a glance. It displays data of obj ectsand other web content in the application.

Activity 1: Creating a Tab f or Customer Obj ect

Click Home, type Tabsin Quick Find, and select Tabs.

Under Custom Obj ect Tabs, click New.

For Obj ect, select Customer.

For Tab Style, choose any icon.

Leave all def ault settingsasthey are.

Click Next, Next, and Save.

Activity 2: Creating a Tab f or Shipment Obj ect

Click Home, type Tabsin Quick Find, and select Tabs.

Under Custom Obj ect Tabs, click New.

For Obj ect, select Shipment.

Choose any Tab Style.

Keep def aultsunchanged.

Click Next, Next, and Save.

Activity 3: Creating a Tab f or Delivery Agent Obj ect

Click Home, type Tabsin Quick Find, and select Tabs.

Under Custom Obj ect Tabs, click New.

For Obj ect, select Delivery Agent.

Pick any Tab Style.

Leave def ault settings.

Click Next, Next, and Save.

Activity 4: Creating a Tab f or Branch Obj ect

Click Home, type Tabsin Quick Find, and select Tabs.

Under Custom Obj ect Tabs, click New.

For Obj ect, select Branch.

Choose any Tab Style.

Keep all def aults.

Click Next, Next, and Save.

Activity 5: Creating a Tab f or Delivery Update Obj ect

Click Home, type Tabsin Quick Find, and select Tabs.

Under Custom Obj ect Tabs, click New.

For Obj ect, select Delivery Update.

Select any Tab Style.

Keep def ault configurations.

Click Next, Next, and Save.

Activity 6: Creating a Tab f or I nvoice Obj ect

Click Home, type Tabsin Quick Find, and select Tabs.

Under Custom Obj ect Tabs, click New.

For Obj ect, select I nvoice.

Select any Tab Style.

Leave def aultsunchanged.

Click Next, Next, and Save.

Milestone 4 : Fields& Relationships

What are fields?
Fields in Salesforce represent what the columns represent in relational databases. I t can store
data valueswhich are required f or a particular obj ect in a record.

Activity 1: Creation of fieldsf or the Customer Obj ect

Field Name	API Name	Data Type	Required	Description
Customer Name	Name	Text	Yes	Full name of customer
Email	Email_ _ c	Email	Yes	Customer

				email
Phone	Phone_ _ c	Phone	Yes	Customer contact number
Address	Address_ _ c	Text Area	Yes	Customer address
Customer Type	Customer_Type_ _ c	Picklist	Yes	Regular / Business
City	City_ _ c	Text	No	Customer cit
State	State_ _ c	Text	No	Customer state
Pincode	Pincode_ _ c	Text	No	Postal code
I sActive	I s_ Active_ _ c	Checkbox	No	Active status

Field Name
API Name
Data Type
Required
Description
Customer Name
Name
Text
Yes
Full name of customer
Email
Email_ _ c
Email
Yes
Customer email
Phone
Phone_ _ c
Phone
Yes
Customer contact number
Address
Address_ _ c
Text Area
Yes
Customer address
Customer Type
Customer_ Type_ _ c
Picklist
Yes

Field Label: Pincode

Save

8. I sActive (Checkbox)
Setup Customer_ _ c Fields New

Checkbox Next

Field Label: I sActive

Save

Activity 2: Creation of fieldsf or the Shipment Obj ect

Field Name	API Name	Data Type	Required	Description
Tracking Number	Name	Auto Number	Yes	Unique I D
Customer	Customer_ _ c	Lookup(Custo mer)	Yes	Related customer
Source Address	Source_Addr ess_ _ c	Text Area	Yes	Pickup location
Destination Address	Destination_ Address_ _ c	Text Area	Yes	Delivery location
Weight	Weight_ _ c	Number	Yes	Parcel weigł
Price	Price_ _ c	Currency	Yes	Shipping charge
Status	Status_ _ c	Picklist	Yes	Shipment status
Expected Date	Expected_Dat e_ _ c	Date	Yes	Delivery estimate
Assigned Agent	Assigned_Age nt_ _ c	Lookup(Delive ry_ Agent)	No	Delivery agent
Shipment Type	Shipment_Ty pe_ _ c	Picklist	Yes	Domestic/ Internationa l

Field Name
API Name
Data Type
Required
Description
Tracking Number
Name

2:46
Courier_Management...

Agent)
No
Delivery agent
Shipment Type
Shipment_ Type_ _ c

Picklist
Yes
Domestic/ I nternational

Favorite
To PDF
Open With
Share

Error Message: Pincode must be 6 digits.

Location: Field Pincode

Save

Activity 2: Create Validation Ruleson Shipment_ _ c

1. Validation Rule: Weight Must Be Greater Than Zero

Steps:

Setup Obj ect Manager Shipment_ _ c

Click Validation Rules New

Rule Name: Validate_ Weight

Active Checked

Formula:

Weight_ _ c <= 0

Check Syntax

Error Message: Weight must be greater than zero.

Location: Field Weight

Save

2. Validation Rule: Price Must Be Greater Than Zero

Steps:

Setup Obj ect Manager Shipment_ _ c

Validation Rules New

Rule Name: Validate_ Price

Active Yes

Formula:

Price_ _ c <= 0

Error Message: Price must be greater than zero.

Location: Field Price

Save

3. Validation Rule: Expected Date Cannot Be in the Past

Steps:

Setup Obj ect Manager Shipment_ _ c

Validation Rules New

Error Message: Pincode must be 6 digits.

Location: Field Pincode

Save

Activity 2: Create Validation Ruleson Shipment_ _ c

1. Validation Rule: Weight Must Be Greater Than Zero

Steps:

Setup Obj ect Manager Shipment_ _ c

Click Validation Rules New

Rule Name: Validate_ Weight

Active Checked

Formula:

Weight_ _ c <= 0

Check Syntax

Error Message: Weight must be greater than zero.

Location: Field Weight

Save

2. Validation Rule: Price Must Be Greater Than Zero

Steps:

Setup Obj ect Manager Shipment_ _ c

Validation Rules New

Rule Name: Validate_ Price

Active Yes

Formula:

Price_ _ c <= 0

Error Message: Price must be greater than zero.

Location: Field Price

Save

3. Validation Rule: Expected Date Cannot Be in the Past

Steps:

Setup Obj ect Manager Shipment_ _ c

Validation Rules New

Rule Name: Validate_ Expected_ Date

Formula:
Expected_ Date_ _ c < TODAY()
Check Syntax

Error: Expected delivery date cannot be in the past.

Location: Field Expected Date

Save

Activity 3: Create Validation Ruleson Delivery_ Agent_ _ c

1. Validation Rule: Email Must Be Valid
Steps:
Setup Obj ect Manager Delivery_ Agent_ _ c

Validation Rules New

Rule Name: Validate_ Agent_ Email

Active Yes

Formula:
NOT(CONTAI NS(Email_ _ c, " @"))
Error Message: Enter a valid email address.

Location: Field Email

Save

Activity 4: Create Validation Ruleson Branch_ _ c

1. Validation Rule: Branch Code Must Not Be Blank
Steps:
Setup Obj ect Manager Branch_ _ c

Validation Rules New

Rule Name: Branch_ Code_ Required

Formula:
I SBLANK(Branch_ Code_ _ c)
Error Message: Branch Code ismandatory.

Location: Field Branch Code

Save

Activity 5: Create Validation Ruleson Delivery_ Update_ _ c

1. Validation Rule: RemarksRequired I f StatusisDelayed

Steps:

Setup Obj ect Manager Delivery_ Update_ _ c

Validation Rules New

Rule Name: Remarks_ Required_ For_ Delay

Formula:

```
AND(
I SPI CKVAL(Status_ _ c, "
Delayed" ),
I SBLANK(Remarks_ _ c)
)
```

Error Message: Remarksare required when statusisDelayed.

Location: Field Remarks

Save

Milestone 6 - Page Layouts:

Page Layouts in Salesforce control what a user sees on a record page and how inf ormation isorganized.

They help structure fields, sections, buttons, related lists, and other componentsin a clean and user- f riendly way.

By customizing page layouts, we ensure that users can easily enter, update, and viewthe required data.

Activity 1: Edit Page Layout f or Customer_ _ c

Steps:

Setup Obj ect Manager Customer_ _ c

Click Page Layouts

Click the def ault layout (usually Customer Layout)

Sectionsto Add & Arrange:

Open the def ault layout

I n the top palette, click Section

Drag the Section and drop it anywhere on the layout

A pop- up appears Fill details:

Section Name: Customer I nf ormation

Layout:

1- Column or

2- Column (recommended)

Tab- Key Order: Top- Down

Click OK

The newsection will appear on the layout.

Drop fieldsinside the section (it will highlight).

Repeat the same f or each field.

Section 1: Customer I nf ormation

Customer Name

Customer Type

Email

Phone

I sActive

Section 2: AddressDetails

Address

City

State

Pincode

Edit the layout asneeded

Click Save.

Activity 2: Edit Page Layout f or Shipment_ _ c

Steps:

Setup Obj ect Manager Shipment_ _ c

Click Page Layouts

Open the def ault Shipment Layout

Sectionsto Add:

Section 1: Shipment Details

Tracking Number

Customer

Shipment Type

Weight

Price

Status

Section 2: AddressI nf ormation

Source Address

Destination Address

Section 3: Delivery I nf ormation

Expected Date

Assigned Agent

Arrange fieldsproperly

Click Save

Activity 3: Edit Page Layout f or Delivery_ Agent_ _ c

Steps:

Setup Obj ect Manager Delivery_ Agent_ _ c

Page Layouts

Open def ault Delivery Agent Layout

Arrange Fields:

Agent Details

Agent Name

Email

Phone

Branch

Status

Joining Date

Additional I nf ormation

Related Shipments

Related Delivery Updates

Edit Save

Activity 4: Edit Page Layout f or Branch_ _ c

Steps:

Setup Obj ect Manager Branch_ _ c

Page Layouts

Open def ault Branch Layout

Sections:

Branch I nf ormation

Branch Name

Branch Code

City

State

Manager Name

Arrange fields

Click Save

Activity 5: Edit Page Layout f or Delivery_ Update_ _ c

Steps:

Setup Obj ect Manager Delivery_ Update_ _ c

Page Layouts

Open def ault Delivery Update Layout

Sections:

Update I nf ormation

Update Number

Shipment

Status

Update Date

Remarks

Arrange properly

Click Save

Activity 6: Edit Page Layout f or I nvoice_ _ c